

ANALYTICAL SOLUTIONS • DENNIS G. ZILL & DR. HAGHIGHAT LECTURE 3

Chapter 2.2: Separable Differential Equations

Quarantining variables, integration protocols, constant equilibrium solutions, and identifying lost singular solutions.

Zill §2.2 (pp. 71–78)

Lecture 3 Slides 3–7

Midterm 1 Core Topic

PLAIN-ENGLISH INTUITION

An equation is **separable** when variables can be cleanly partitioned to opposite sides of the equals sign: all y 's bundled with dy , and all x 's bundled with dx . Once quarantined, you integrate both sides independently. But beware: **dividing by $h(y)$ deletes constant solutions!** Always find roots $h(y) = 0$ before you divide!

STANDARD FORM & DEFINITION

A first-order DE is **separable** if $\frac{dy}{dx}$ factors into a product:

$$\frac{dy}{dx} = g(x)h(y) \iff \frac{1}{h(y)} dy = g(x) dx$$

Integrating both sides yields: $\int \frac{1}{h(y)} dy = \int g(x) dx + C$.

THE 4-STEP SEPARATION PROTOCOL

1. **Roots Check:** Find roots of $h(y) = 0$. Each root $y = r$ is an equilibrium constant solution!
2. **Separate:** Rearrange into $\frac{1}{h(y)} dy = g(x) dx$.
3. **Integrate:** Evaluate antiderivatives on both sides.
4. **Solve Explicitly:** Isolate $y(x)$ if possible. Verify if any constant root from Step 1 was lost!

THE LOST SINGULAR SOLUTION HAZARD

When you divide by $h(y)$, you assume $h(y) \neq 0$. If $y = r$ makes $h(r) = 0$, but no finite value of the constant C can produce $y = r$ from the general solution formula, $y = r$ is a **singular solution!** You must write it explicitly to get full marks on exams.

Master Worked Example 1 • Circle & Branch Selection (Lecture 3 Slide 5)

CONCORDIA LECTURE

Problem: Solve the IVP: $\frac{dy}{dx} = -\frac{x}{y}$, with $y(4) = -3$. State the domain of definition.

1 Separate Variables:

$$y dy = -x dx \implies \int y dy = \int -x dx + C_1.$$

$$\frac{1}{2}y^2 = -\frac{1}{2}x^2 + C_1 \implies x^2 + y^2 = C \text{ (where } C = 2C_1\text{)}.$$

This represents concentric circles centered at the origin!

2 Apply IVP & Select Branch:

$$(4)^2 + (-3)^2 = 16 + 9 = 25 \implies C = 25.$$

$$\text{Implicit curve: } x^2 + y^2 = 25.$$

$$\text{Solving explicitly: } y = \pm\sqrt{25 - x^2}.$$

Since $y(4) = -3 < 0$, we MUST pick the **negative branch**:

$$y(x) = -\sqrt{25 - x^2} \text{ on } x \in (-5, 5)$$

2 Partial Fractions, Logarithm Rules & Singular Solutions

Master Worked Example 2 · The Textbook Classic (Lecture 3 Slide 6 & Zill Ex 3)

STEP-BY-STEP BREAKDOWN

Problem: Solve the differential equation $\frac{dy}{dx} = y^2 - 4$, with initial condition $y(0) = 0$. Identify all singular solutions.

1 Find Constant Solutions (Before Dividing!):

$$y^2 - 4 = 0 \implies (y - 2)(y + 2) = 0 \implies y = 2 \text{ and } y = -2 \text{ are constant equilibrium solutions.}$$

2 Separate Variables & Partial Fractions:

$$\frac{dy}{(y-2)(y+2)} = dx.$$

$$\text{Decompose: } \frac{1}{(y-2)(y+2)} = \frac{A}{y-2} + \frac{B}{y+2}.$$

$$1 = A(y+2) + B(y-2).$$

$$\bullet \text{ Let } y = 2 \implies 1 = 4A \implies A = 1/4.$$

$$\bullet \text{ Let } y = -2 \implies 1 = -4B \implies B = -1/4.$$

$$\frac{1}{4} \left(\frac{1}{y-2} - \frac{1}{y+2} \right) dy = dx$$

3 Integrate Both Sides:

$$\frac{1}{4} \ln |y-2| - \frac{1}{4} \ln |y+2| = x + c_1.$$

$$\ln \left| \frac{y-2}{y+2} \right| = 4x + 4c_1.$$

4 Exponentiate & Isolate $y(x)$:

$$\left| \frac{y-2}{y+2} \right| = e^{4c_1} e^{4x} \implies \frac{y-2}{y+2} = C e^{4x} \quad (C = \pm e^{4c_1}).$$

$$y - 2 = C e^{4x} (y + 2) \implies y(1 - C e^{4x}) = 2(1 + C e^{4x}).$$

$$y(x) = 2 \frac{1 + C e^{4x}}{1 - C e^{4x}}$$

5 Apply Initial Condition $y(0) = 0$:

$$0 = 2 \frac{1+C}{1-C} \implies 1+C = 0 \implies C = -1.$$

$$\text{Particular Solution: } y(x) = 2 \frac{1-e^{4x}}{1+e^{4x}} = -2 \tanh(2x).$$

6 Singular Solution Check:

$$\bullet \text{ If } C = 0 \implies y = 2 \text{ (recovered!).}$$

$$\bullet \text{ But } y = -2 \text{ requires } C \rightarrow \infty \text{ (cannot be produced by any finite real } C).$$

$$\implies y = -2 \text{ is a lost singular solution!}$$

Worked Example 3 · Exponential Separation

RAPID SEPARATION

Problem: Solve $\frac{dy}{dx} = e^{3x+2y}$.

$$\bullet \text{ Factor exponent: } \frac{dy}{dx} = e^{3x} e^{2y} \implies e^{-2y} dy = e^{3x} dx.$$

$$\bullet \text{ Integrate: } -\frac{1}{2} e^{-2y} = \frac{1}{3} e^{3x} + C_1 \implies e^{-2y} = C - \frac{2}{3} e^{3x}.$$

$$\bullet \text{ Take logarithm: } -2y = \ln \left(C - \frac{2}{3} e^{3x} \right) \implies y(x) = -\frac{1}{2} \ln \left(C - \frac{2}{3} e^{3x} \right).$$